

TCS349: Responsible and Explainable AI

Unit 2, Lecture 9

Balancing Security & Privacy: Unit II Case Study & Review

Where security and privacy pull in the same direction, where they conflict, and how transparency, accountability, surveillance and privacy come together in practice.

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Security



Privacy



Balancing Both



Unit II Review

1. The Security-Privacy Tension in AI

Security aims to protect systems and data from unauthorised access, misuse or attack.

Privacy aims to protect individuals from unwanted collection, use or exposure of their personal data.

The two goals often require different, sometimes opposing, technical choices.

Responsible AI treats this as a tension to be actively managed, not a problem with a single correct answer.

2. Where Security and Privacy Align

Access control, covered in Lecture 17, serves both goals at once: it protects data from attackers and limits who can see personal information.

Encryption protects data from unauthorised access while also reducing the privacy harm if a breach occurs.

Audit logs, from Lecture 12, support both security monitoring and privacy accountability using the same record.

In these cases, investing in one goal directly strengthens the other.

3. Where Security and Privacy Conflict

Strong security monitoring often requires collecting more data - such as detailed logs and behavioural signals - which increases privacy exposure.

Anonymisation, which protects privacy, can remove information a security team needs to detect and investigate an attack.

Facial recognition for building security directly trades individual privacy for a security benefit, as seen in Lecture 15.

Neither goal should be allowed to fully override the other without a deliberate, justified decision.

4. A Problem-Solving Framework for Balancing Both

Identify the specific security risk and the specific privacy risk in the situation, rather than treating it as a vague trade-off.

Ask whether a technique exists that serves both goals at once, such as access control or encryption, before assuming a trade-off is necessary.

Where a genuine trade-off remains, apply the least privacy-invasive option that still achieves the required security outcome.

Document the decision and its reasoning, using the transparency and accountability practices from earlier in this unit.

5. Unit II Case Study - Setup

A city proposes installing AI-powered cameras with facial recognition and behavioural monitoring in public transit stations to reduce crime.

The system would identify persons of interest, track unusual movement patterns, and retain footage for 90 days.

Civil liberties groups raise concerns about surveillance, profiling and a lack of clear oversight.

The transit authority must decide how, or whether, to deploy the system responsibly.

6. Unit II Case Study - Applying the Concepts

Transparency (Lectures 11-12): the authority should publish a system card describing the technology's purpose, accuracy and limitations.

Accountability (Lectures 13-14): a named oversight body should review matches and handle escalations, not the system alone.

Privacy (Lectures 16-17): footage should be minimised and anonymised wherever possible, with strict access control and a limited retention period.

Balancing security and privacy: facial recognition could be restricted to a documented list of serious offences, not open-ended monitoring.

7. The Debate: Facial Recognition in Public Spaces

Arguments For	Arguments Against
Can deter and help solve serious crime in high-risk public areas	Enables constant monitoring of people who have done nothing wrong
Frees officers from manually reviewing large volumes of footage	Documented accuracy gaps disproportionately misidentify some groups
Can be scoped narrowly to specific, serious offences	Systems deployed narrowly often expand in scope over time

8. Unit II Review - Key Concepts

Transparency: making an AI system's data, logic and limitations understandable to relevant stakeholders.

Accountability: assigning clear responsibility, human oversight and escalation paths for a system's decisions.

Surveillance: AI techniques - facial recognition, video analytics, behavioural monitoring - that observe people at scale, each with distinct risks.

Privacy: protecting personal data through consent, purpose limitation, minimisation, anonymisation and access control.